ARYAN GAUTAM JADHAV

AI & Machine Learning Engineer

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PROFILE

AI Researcher & M.Sc. AI & Robotics student with a foundation in Mechanical Engineering. Brings hands-on experience deploying and maintaining production ready ML models at scale and holds a published patent in IoT-ML integration for Industry 4.0. Seeking research / engineering roles or internships in AI, machine learning or robotics.

WORK EXPERIENCE

Machine Learning Engineer **06/2023 - 03/2024**
AI Variant | Pune, India

- Built and deployed 5+ ML models (SVM, Random Forest, ANN, CNN, RNN) containerised via Docker, achieving >95% accuracy across classification and regression tasks.
- Designed 3 end-to-end recommendation systems using collaborative and content-based filtering, reducing manual curation effort by 40%.
- Developed and optimised Python-based ML pipelines using Newton & quasi-Newton methods and clustering algorithms (K-Means, DBSCAN, Agglomerative); managed versioning via Git.
- Delivered a resume classifier (ANN, 96% accuracy) containerised and adopted by the company for live recruitment screening.

Management Assistant (Internship) **01/2022 - 03/2022**
Nickunj Group of Companies | Pimpri-Chinchwad, India

- Assisted in coordinating procurement workflows and vendor communications across seven product lines.

EDUCATION

M.Sc. Artificial Intelligence and Robotics **03/2025 - Present**
Hochschule Hof | Hof, Germany

Modules: Deep Learning, Intelligent Robotics, Advanced Architectures in AI, Gen AI, Neuromorphic AI, Predictive Maintenance, Data Analysis & ML, Explainable AI (XAI), Industry 4.0

Bachelor of Engineering (Mechanical) with Honours in Robotics **07/2019 - 07/2023**
Savitribai Phule Pune University | Baramati, India

SKILLS

ML / DL Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, snnTorch, XGBoost
LLMs & GenAI: HuggingFace Transformers, LangChain, OpenAI/ Gemini APIs, RAG, FAISS
Languages & Tools: Python, SQL, Git, NumPy, Pandas, Matplotlib, Seaborn
MLOps & Deployment: Docker, FastAPI, MLflow, GitHub, Streamlit, Kubernetes, Roboflow
Robotics & Vision: ROS2, Gazebo, OpenCV, PCL, LiDAR, YOLO
Specialisations: Neuromorphic Computing, Computer Vision, NLP, Deep Learning, Recommendation Systems, Time Series
Cloud & Platforms: AWS (basics), GCP Vertex AI, Jupyter, PyCharm, VS Code

PROJECTS

Neuromorphic Computing & SNN Benchmarking **10/2025 - 03/2026**

Engineered a deep Spiking Neural Network (SNN) using PyTorch and LIF neurons via snnTorch, achieving 98.33% accuracy surpassing the static ANN baseline of 97.39%. Demonstrated 96.7% temporal sparsity and a 3.17× reduction in theoretical energy consumption (2,477 nJ vs. 7,864 nJ per inference).

Genesis v1.0 - Conversational AI Chatbot **04/2025 - 10/2025**

Built a Streamlit-based context-aware chatbot using the Google Gemini API with prompt-engineered instructions for specificity and accuracy. Deployed as a live web application.

Autonomous Navigation Simulation **03/2025 - 07/2025**

Simulated autonomous robot navigation using ROS/Gazebo with LiDAR data processing. Developed real-time obstacle avoidance for static and dynamic objects and implemented path planning algorithms to reach defined destinations.

Driver Drowsiness Detection System **05/2024 - 10/2024**

Real-time drowsiness detector with 98% accuracy using eye aspect ratio (EAR) monitoring via an input camera; triggers visual and audio alerts when eyes are closed beyond a set threshold.

Drone Detection System - YOLOv8 **12/2023 - 05/2024**

Fine-tuned a YOLOv8-based model via Roboflow, achieving 90.4% accuracy for real-time drone detection across images, video, and live streams.

Intelligent Resume Classifier (ANN) **11/2023 - 04/2024**

ANN-based resume classification model with 96% accuracy targeting domain-specific keywords; adopted by AI Variant for live recruitment screening.

Personalized Recommendation Engine **08/2023 - 11/2023**

Neural network-based recommendation system analysing historical data, genre, plot, and cast to suggest titles. Achieved a Precision@10 of 88% and an NDCG score of 0.85, significantly outperforming baseline collaborative filtering methods.

Energy Production Predictor **11/2022 - 02/2023**

Linear regression model with 98% accuracy predicting future power plant energy generation from historical data.

HONOURS & AWARDS

Patent: Integrating IoT and Machine Learning Revolution: The driving force of Industry 4.0 **08/2022**

IP India Journal

Developed a closed-loop industrial framework integrating IoT sensor networks with ML models for real-time anomaly detection, predictive maintenance, and automated process optimisation in Industry 4.0 environments.

Gold Medalist: Product Design & Development **03/2023**

Indian Institute of Technology Roorkee | NPTEL

Secured 90% marks in the PDD examination organised by IIT Roorkee in collaboration with NPTEL; awarded Gold + Elite certification.

LANGUAGES

- **English:** C2 (Proficient)
- **German:** B1 (Intermediate)
- **Hindi:** Native Proficiency
- **Marathi:** Mother Tongue

INTERESTS

- Basketball, Fencing
- Formula 1 (Data Analysis & Strategy)
- Cooking
- Embedded Systems